

```
1 clc;
2 clear;
3 q = 5*10^-6;
4 r = 3*10^-3;
5 epsilon = 8.854*10^-12;
6 pi = %pi;
7 E = q/(4*pi*epsilon*r^2);
8
9 disp("Electric Field Intensity (E) in V/m =");
10 disp(E);
11 D = epsilon * E;
12 disp("Electric-Flux-Density (D) in C/m^2 =");
13 disp(D);
14
15
16 x = linspace(0,10,50);
17
18 y = E * exp(-0.1*x);
19 z = D * exp(-0.1*x);
20
21 figure();
22
23 subplot(2,2,1);
24 plot2d3(x,y);
25 xlabel("Distance");
26 ylabel("Electric Field (V/m)");
27 title("Electric-Field-Intensity");
28
29 subplot(2,2,2);
30 plot2d3(x,z);
31 xlabel("Distance");
32 ylabel("Electric Flux Density (C/m^2)");
33 title("Electric-Flux-Density");
34
```

Scilab 2025.1.0 Console

File Edit Control Applications ?

Scilab 2025.1.0 Console

```
"Electric Field Intensity (E) in V/m ="
4.993D+09
"Electric Flux Density (D) in C/m^2 ="
0.0442097
```

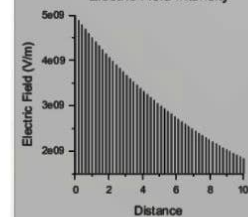
-->

Graphic window number 0

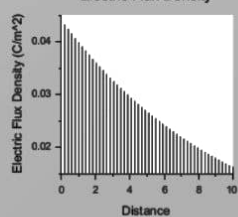
File Tools Edit ?

Graphic window number 0

Electric Field Intensity



Electric Flux Density



Exp: 2

$$Q = 5 \times 10^{-6}$$

$$r = 3 \text{ mm}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$= \frac{5 \times 10^{-6}}{4 \times 3.14 \times 8.854 \times 10^{-12} \times (3 \times 10^{-3})^2}$$

$$r^2 = (3 \times 10^{-3})^2 = 9 \times 10^{-6}$$

$$4\pi\epsilon_0 r^2 = 12.5664 \times 8.854 \times 10^{-12} \times 9 \times 10^{-6}$$
$$= 1.0018 \times 10^{-15}$$

$$E = \frac{5 \times 10^{-6}}{1.0018 \times 10^{-15}}$$

$$E = 4.99 \times 10^9 \text{ V/m}$$

Electric flux

$$D = \epsilon_0 E$$

$$(8.854 \times 10^{-12}) (4.99 \times 10^9)$$

$$D = 0.0442 \text{ C/m}^2$$